

Esame: 19 giugno ore 11 aula seminari. Portare CDI, ricevuta prenotazione, e due copie della tesina con parole chiave stampate.

Today, we deal with the AI act. It deals with 80 articles.

The story that led to the approval of the AI act is an example of an attempt to produce a democratic law within a democratic context.

The first proposal was made in 2021, and the full text was made in 2024, and will fully enter in force in August 2026.

From the moment this act was conceived, until it is being enforced, an evolution has taken place, and the law, in order to keep up, must certainly be rigorous in one hand, while in the other it should be flexible to adapt.

The difficulties this act encountered are countless:

It's the first legislative act to regulate new technology. Europe, being the first to legislate in this sense, has been strategically positioned as a driving force, compared to other international competitors.

USA invents, China produces, and Europe legislates (this is a saying).

It's also said that Europe is the "culla" of justice.

If countries don't want to be cut out by the European market must comply with the rules of the AI act.

The AI act applies whenever a technology is given to European users.

A long and very harsh negotiation process involved many actors, academics, technicians and experts. There were sessions lasting 40 hours of debate, trying to discuss and reach an agreement.

The purpose of the AI act is the creation of a set of rules shared by European states, capable of ensuring the creation and use of AI systems, that are anthropocentric, so safe and respectful of fundamental rights, democratic principles, and the rule of law.

Europe therefore intended to provide a uniform legal framework for the states, in order to prevent market fragmentation, reducing the legal uncertainty among the different states.

How can these goals be achieved? These acts intend to promote a responsible use of these digital tools, offering a series of protection for citizens, against manipulative use of AI, creating the possibility of bringing out legal action caused by abuses of these technologies.

While on one hand this law wants to put a limitation on large companies, in the other it intends to promote small companies and start-ups.

Moreover, in addition to these ambitious objectives, we also have environmental

sustainability. We know that behind the appearance of clean and lightness the digital world is heavily material. There are chemical agents used to produce components, etc., and framing all this are the antisocial behaviours, especially in poor countries with weak unionization.

There are kilometers of cable, and destructive work to bury these cables underground and put them under the sea.

There is also the enormous quantity of energy consumed by machines.

It's difficult to ensure a balanced approach between ecology and economy, and often this ends up justifying the environmental exposure.

If we want to produce more bread, it's not sufficient to create always more sophisticated and technology advanced ovens, we also need wheat and flour to do it. It's a balance we should take into account.

We will see what solution will be adapted in the future, and whether it will be possible.

Let's talk about the structure of the AI act. It's a regulation, which is a legal act that automatically applies to all member states in a uniform manner. Its peculiarity is that, unlike directives, it does not need further approval to be valid. Unlike the universal declaration of human rights, which is just a declaration, and so it only possesses guiding powers, the AI act establishes general rules that are binding.

This act had a long incubation, and in order to arrive at a full text, recurring to the "trilogue" in 2023. It's an informal inter-institutional ... that brings together representatives of the parliament and ...

The purpose is to reach a provisional agreement shared both by parliament and counselors (??), which are the bodies of the European Union.

The rules contained within the AI act will become fully operational in August, and until then companies are encouraged to sign the AI pact, that is to voluntarily adhere to these rules, in order to comply with the rules that will be established. Since for companies aligning requires substantial investment for implementing practices, an attempt was made to balance this need with the objective of encouraging small realities, through excessive regulations regarding only large companies.

What kind of approach was adapted to organize this discipline? The attempt was to reconcile the parties ... with the potential risk the use may entail.

The AI act classifies and regulates AI by adopting a risk-based approach, distinguishing between unacceptable risk, high risk, medium risk and negligible risk.

Who's in charge of the risk analysis? Developers and so technicians, this is one of the cases where technicians become legislators, as based on the definition technicians give to one system or another, completely different regulations are ???. Evaluating the risk of applying technology in a certain field is different from evaluating the risk the use of the technology entails in itself. This is a precise political choice.

Using another approach, that indicates risk within the technology itself, would have given a more exhaustive but less flexible regulatory framework, but the AI would have never come to life perhaps.

One of the greatest difficulties in drafting the act arose in establishing the definitions of context.

We're dealing with 27 member states that must state different interpretations to converge, deriving from the country's history. All of this should converge within a shared agreement.

Much work has been done on prohibited applications, mainly those who involved unacceptable risks.

Among these prohibited practices, there's the biometric categorisation of sensitive data, like political or religious beliefs, ethnic....

This is aimed at avoiding discriminatory practices linked to identification.

Here we see how the possible drift is reducing the human being to its data.

Untargeted scraping of facial images, images found online cannot be collected to create databases without specific objectives. The system that was intended to protect privacy, may turn in a system that will not.

Another prohibition on the use of AI concerned emotion recognition in work place and educational context. Although machines do not directly experience emotions, are very competent at recognising them, and this could lead to a violation of human freedom.

Social scoring is also prohibited. This system is for evaluating citizens' behaviours based on online and offline parameters.

It was like a point-based driver license system, and when people would not maintain a score, people would be excluded from social participations, up to sanctions including imprisonment.

Our data, interactions, could be interpreted through a narrative lens that contributes to build the social profile of an individual.

Some technologies can make the system particularly powerful, if we consider that it combines neuroscience, storytelling and the psychology of perception.

Manipulative techniques that exploit vulnerabilities are also prohibited.

Among practices we have predicting and real time biometric recognition.

Parliament tried to oppose a strong resistance to these practices.

Exceptions to the prohibition of the use were established, permitted in:

1. Imminent terrorist threat, so for public security
2. Criminal investigation to search for perpetrators suspect and victims of serious crimes.
3. Persecution of serious crimes

The attempt to confine these practices to specific case must confront with the opacity of the definition, and the ambiguity of language (what do we mean exactly with "serious"?).

We can all be suspect. The prohibition of predictive policy was also the subject of hard negotiation, and consists in a set of activities aimed at anticipating crimes through the combination of data related to reports of previously committed crimes. A great achievement of the parliament was preventing predictive policies from targeting individual purposes, as now it can be used to prevent crimes, but operating on anonymous data.

This is also dangerous though.

In any case, it was possible to bind the use of this technology by law enforcement to a series of safe guards, like notifying to independent authorities of the use of these AI systems, trying to guarantee transparency in government sectors.

The use of these technologies by the executive power must be authorised ex-ante, or in urgent cases 24h in advance.

The difficulties in organisation between different states demonstrated the reluctance to delegate national security to Europe.

There are countries adopting a more harsh approach towards terrorism, and thus tends to approve a specific use of the AI.

The attempt to place central issues revealed the national ideologies animating different European sensitivities, strengthening the nationalism.

As regard AI risk, we include healthcare, transport and public security.

These systems are required to meet high standards of safety and reliability.

Companies must ensure that these systems have been examined, and must be controlled by human supervision.

The rule of human supervision is aimed by containing the risks aimed by AI, which could impact human rights.

Example: Petrov, a Russian officer, was tasked with monitoring the rapid nuclear warning system protecting the Soviet airspace. His job was to immediately alert the superiors if the computer gave any indication of an American attack. On 26th

septembre 1993 Petrov was on duty, where after midnight sirens began to sound. According to pc, an enemy missile was headed towards the territories.

There was something that did not convince Russian officer, the fact the algorithm only have seen 5 missiles, so Petrov choose to wait, and 23 minutes later when it was clear that no missile had strucked russian soil, he realised that he was right. If he reported the alarm, a nuclear conflict would have started.

This example shows the human factor appears to be a crucial element in decision making process, and giving a person the opportunity to examine an algorithm suggestion is the only way to avoid problems.

The only problem is that not even humans are always reliable, as shown by the case of the smiler roller coaster, where two engineers called to carry out a....

Therefore, given both algorithmic and human fallibility, who should have the final say? This is an open question.

System must also provide post-approval monitoring to verify long term compliants with rules and fundamental rights.

Companies using high risk AI are subject to strict regulations, and in particular we refer to systems ..., like a deepfake showing Trump arrested. The news spread rapidly, causing big consequences on markets and elections.

This policy pursue an approach to mitigate risks deriving from unregulated use of technologies.

As regards tech like chatbots, transparency rules are much less transparent, limited to informing users.

Technologies considered minimal risk like filters and videogames can enjoy more freedom.

Another area concern LLM, refereed as general purpose AI. These technologies were classified following the american model according to FLOPS, that is accoring to computational capacities and not according to revenues.

GPAI are called high ... systems, and are those with a threshold of FLOPS, which is a gicant computational capacity, possessed only by ChatGPT, Gemini and DeepSeek.

Here we notice a curious aspect of the law, which needs to be general. IN this case, it's clear that is addresses only to two or three operators.

They msut comply with rigouours requirement.

For non-high impact system, the regularoty provisions are limited to imposing compliance with specific obligations, by guaranteeing cyber secrity, rubstenn,.

Robustness means resilience to attacks, bugs, accidents, so the capability to keep

data safe.

As regarded model used for research purpose, it requires minimal regulations.

In this regard, sandboxes have been ..., that is experimental environments

The logic of this provision is first test whether a system works, and then regulate it.

Let's now move to the sanctions framework. We've seen how the timelines for the entry of the enforcing of various rules last over variable amount of months.

Violating rules may result in heavy fine according to seriousness and the size of the company.

The sanctions are economic and range from 35 billion of dollars, of 7% of the annual global turnover, 50 billions of dollars in case the document provided is misleading or missing, and it could reach up to 70 billions.

We must consider a crucial aspect of this issue. If violating certain obligations result in the imposition of large penalties, I (as a company) might not be discouraged in violating this.

Lastly, a very delicate aspect must be highlighted, that is the military use of AI. It does not fall under these rules, continuing to exist under a sort of permissive anarchy.

What would become the impact of AI in human rights? If large companies financing and conducting researches on AI, pursuing profit, can they also pursue the public interest?